

Practice Problems

1. State in words the next elementary row operation that should be performed on the system in order to solve it. [More than one answer is possible in (a).]

$$\begin{aligned} \text{a) } x_1 + 4x_2 - 2x_3 + 8x_4 &= 12 \\ x_2 - 7x_3 + 2x_4 &= -4 \\ 5x_3 - x_4 &= 7 \\ x_3 + 3x_4 &= -5 \end{aligned}$$

Augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 4 & -2 & 8 & 12 \\ 0 & 1 & -7 & 2 & -4 \\ 0 & 0 & 5 & -1 & 7 \\ 0 & 0 & 1 & 3 & -5 \end{array} \right]$$

$$\boxed{\text{Row 4} = \text{Old row 4} + \frac{1}{5} \times \text{Row 3}} \rightarrow \text{One possibility}$$

$$\boxed{\text{Row 3 and Row 4 are interchanged}} \rightarrow \text{One possibility}$$

$$\begin{aligned} \text{b) } x_1 - 3x_2 + 5x_3 - 2x_4 &= 0 \\ x_2 + 8x_3 &= -4 \\ 2x_3 &= 3 \\ x_4 &= 1 \end{aligned}$$

Augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & -3 & 5 & -2 & 0 \\ 0 & 1 & 8 & 0 & -4 \\ 0 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right]$$

The system is already in triangular form

$$\boxed{\text{Row 1} = \text{Old row 1} + 2 \times \text{Row 4}}$$

2. The augmented matrix of a linear system has been transformed by row operations into the form below. Determine if the system is consistent.

$$\left[\begin{array}{cccc|c} 1 & 5 & 2 & +6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 5 & 0 \end{array} \right]$$

$$\begin{aligned} x_1 + 5x_2 + 2x_3 &= -6 \\ 4x_2 - 7x_3 &= 2 \\ 5x_3 &= 0 \end{aligned}$$

$$\therefore x_3 = 0 \quad 4x_2 - 7 \times 0 = 2 \Rightarrow x_2 = \frac{1}{2}$$

$x_1 + 5 \times \frac{1}{2} + 2 \times 0 = -6 \Rightarrow x_1 = -\frac{17}{2}$. $\therefore$ Hence, at least one solution exists. and so the system is consistent.